

AWS- Route 53

VISHWANATH M S

VISHWACLOUDLAB.ORG

Concepts of Route 53

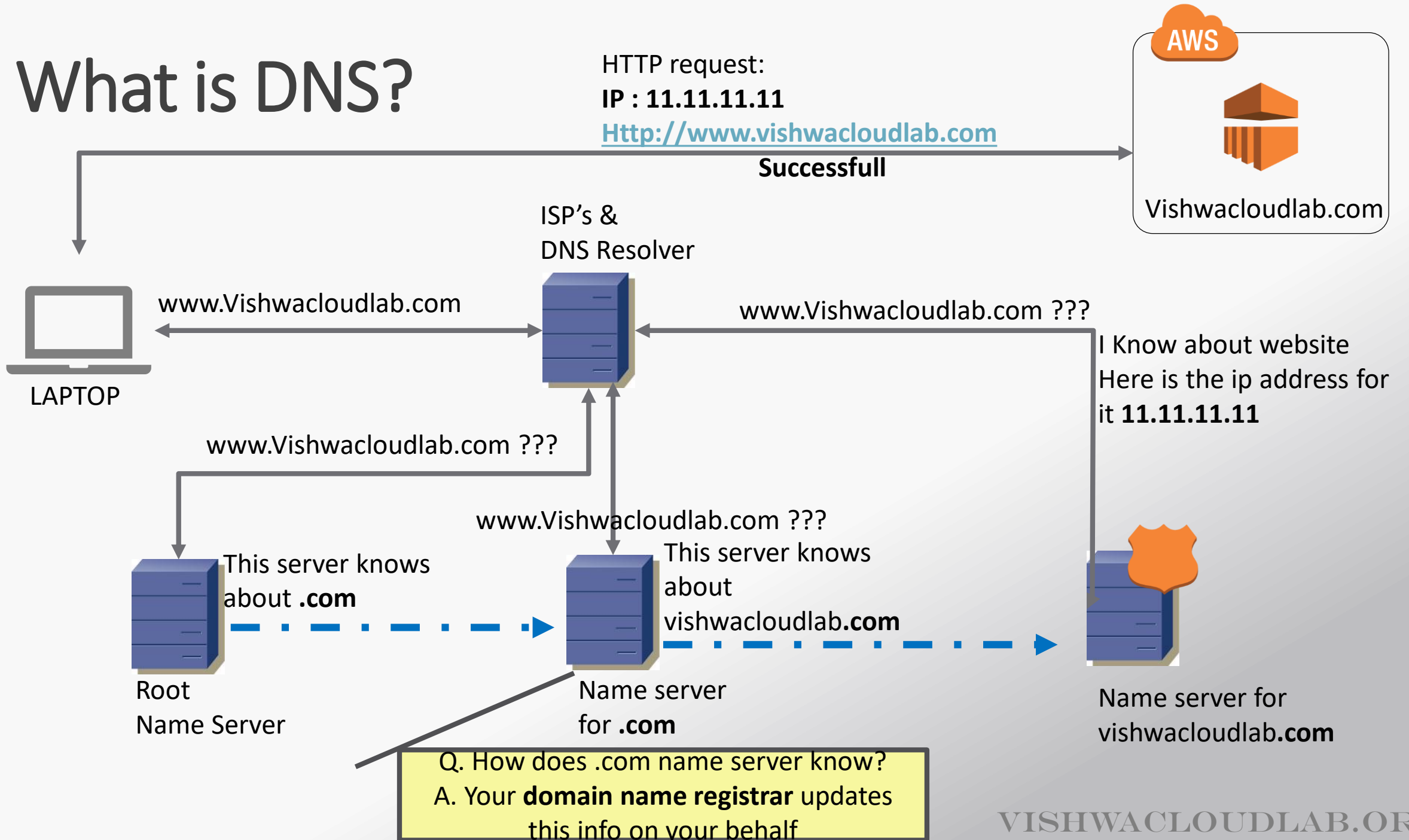
- **Amazon Route 53 is a highly available and scalable Domain Name System (DNS) web service**
- **What is DNS?**
 - The Domain Name System (DNS) is one of the foundations of the internet
 - DNS is a directory of names that match with numbers
 - It is still possible to type in a specific IP address into a browser to reach a website.

How DNS servers Work?

➤ How DNS servers Work?

- The DNS directory that matches name to numbers isn't located all in one place in some dark corner of the internet.
- More than 332 million domain names listed at the end of 2017 and a single directory would be very large indeed to handle the information.
- Each named site can correspond to more than one IP address.
- The Directories are distributed among many servers in the internet
- DNS information is shared among many servers, but is also cached locally on client computers

What is DNS?



Advantages of Managed DNS

- Worldwide anycast network with redundant locations
- 100% availability SLA
- Advanced routing: LBR, GEO, WRR, Failover
- AWS integrations: Alias
- Manage via API, CLI, SDKs, AWS tools, third-party tools

Concepts of Route 53

- Three main functions
 - Register domain names
 - Can register a new Domain name
 - Also, connect to the existing Domain name
 - Route internet traffic to the resources for your domain
 - Check the health of your resources

Record Type

- **A Record** – Domain name to **IPV4** conversion
- **AAAA Record** -- Domain name to **IPV6** conversion
- **C_Name Record** – Domain name to Alias Name

Routing Policy

- **Simple** – Map Domain and Sub Domain name with IP4/IP16
- **Latency** -- LBR works by routing your customers to the AWS endpoint that provides the fastest experience based on actual performance measurements of the different AWS regions where your application is running.
- **C_Name Record** – Domain name to Alias Name

Routing Policy

Simple –

Latency –

Failover –

Geolocation --

Pricing – Health Checks

Health Checks

Get Started With DNS Failover At No Additional Cost*

New and existing customers can create up to 50 health checks for AWS endpoints** that are within or linked to the same AWS account.

	AWS Endpoints	Non-AWS Endpoints
Basic Health Checks	\$0.50* per health check / month	\$0.75 per health check / month
Optional health check features:		
• HTTPS • String Matching • Fast Interval • Latency Measurement	\$1.00 / month per optional feature	\$2.00 / month per optional feature

Troubleshooting Route 53

<https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/troubleshooting-route-53.html>